# Angle facts and polygons

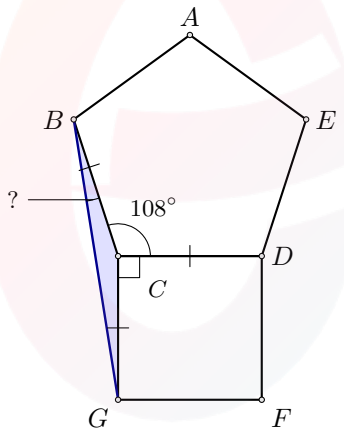
L1-GE1 Level 1 Foundations · Geometry

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How to find angles using lines, parallel lines, triangle and polygon angle sums, isosceles triangles and regular polygons.
**By the end you can** use angles on lines and at points, parallel lines and triangle angle sums, with reasons; find equal sides and use isosceles triangles to calculate unknown angles; work out interior and exterior angles of polygons and decide which regular polygons fit round a point.
**Prerequisites:** none. **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 60 minutes.

## 1 Lesson

The diagram shows a regular pentagon $ABCDE$ and a square $CDFG$ drawn on its side $CD$, outside the pentagon. Find $\angle CBG$.




No angle of triangle $BCG$ is given, so we look for facts that produce one. Each angle of a regular pentagon is $108^\circ$ (proved below) and each angle of a square is $90^\circ$. The angles round $C$ add to $360^\circ$, so $\angle BCG = 360^\circ - 108^\circ - 90^\circ = 162^\circ$. Next we look at lengths: $CB$ is a side of the pentagon and $CG$ is a side of the square, and both equal $CD$. So triangle $BCG$ is isosceles with $CB = CG$. Its base angles are equal and add to $180^\circ - 162^\circ = 18^\circ$, hence $\angle CBG = 9^\circ$.

Most angle questions on the JMC, the Junior Kangaroo and the AMC 8 combine a few facts like these three. This unit proves each one and shows how to choose the next one to use.

### 1.1 Angles at a point and on a line

A full turn is $360^\circ$, so the angles round a point add to $360^\circ$, and angles on one side of a straight line add to $180^\circ$. When two lines cross, the opposite angles are equal; they are called *vertically opposite* angles. Indeed, if the four angles in order are $a$, $b$, $c$, $d$, then $a + b = 180^\circ$ and $b + c = 180^\circ$, since each pair lies on a straight line, so $a = c$.



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## 1.2 Parallel lines

**Fact 1 (parallel lines).**
When a line crosses two parallel lines, corresponding angles are equal (the F shape), alternate angles are equal (the Z shape), and co-interior angles add to $180^\circ$ (the C shape).

Parallel lines point in the same direction, so the crossing line makes the same angles with each of them; angles in matching positions at the two crossings are the corresponding angles. An alternate angle is vertically opposite a corresponding angle, and a co-interior angle is the neighbour of a corresponding angle on a straight line, which gives the other two facts. Each fact also works in reverse: if alternate angles are equal, then the lines are parallel.

## 1.3 Triangles

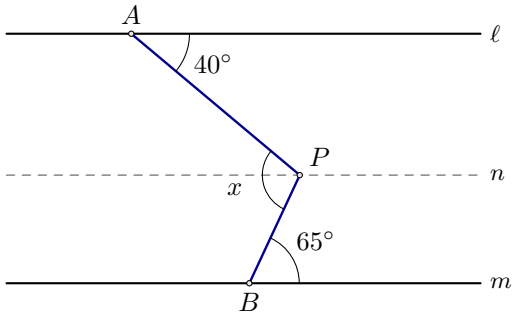
**Fact 2 (triangles).**
The angles of a triangle add to $180^\circ$. An exterior angle of a triangle equals the sum of the interior angles at the other two vertices.

*Proof.* Take a triangle $ABC$ and draw the line through $A$ parallel to $BC$. The angles this line makes with $AB$ and $AC$ are alternate to $\angle B$ and $\angle C$, and together with $\angle A$ they fill a straight line, so $\angle A + \angle B + \angle C = 180^\circ$. The exterior angle at $C$ is the angle between $BC$ and the line $AC$ extended beyond $C$. It lies on a straight line with $\angle C$, so it equals $180^\circ - \angle C$, which is $\angle A + \angle B$.

**Fact 3 (isosceles triangles).**
If two sides of a triangle are equal, then the angles opposite them are equal. Conversely, if two angles are equal, then the sides opposite them are equal.

If $AB = AC$, fold the triangle along the line bisecting $\angle A$. The equal sides $AB$ and $AC$ land on each other, so $B$ lands on $C$ and the triangle is symmetric; hence $\angle B = \angle C$. If the angle between the equal sides (the *apex* angle) is $\theta$, each base angle is $\frac{1}{2}(180^\circ - \theta)$. Each angle of an equilateral triangle is $60^\circ$. (The converse is proved by folding along the perpendicular bisector of $BC$ instead.)

Contest questions rarely say isosceles: equal sides come from one regular polygon, from two shapes that share an edge (as in the opening problem), or from lengths marked equal.



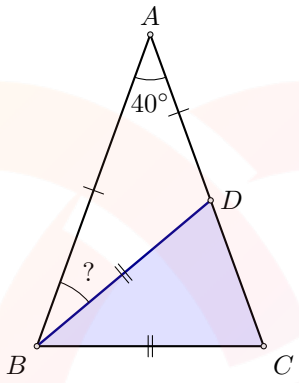

**Example 1.**
Lines $\ell$ and $m$ are parallel. A point $P$ between them is joined to $A$ on $\ell$ and to $B$ on $m$, as



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shown. The angle between $AP$ and $\ell$ is $40°$, and the angle between $BP$ and $m$ is $65°$. Find $x = \angle APB$.

*Solution.*
No triangle contains $x$ together with the given angles, and only Fact 1 links angles at different points on parallel lines. So we create a crossing at $P$: draw the line $n$ through $P$ parallel to $\ell$ and $m$.
The line $n$ splits $x$ into two parts. The upper part and the $40°$ angle are alternate angles between the parallel lines $\ell$ and $n$, so the upper part is $40°$. Similarly the lower part is alternate to the $65°$ angle, so it is $65°$. Hence $x = 40° + 65° = 105°$.
In general, when a path bends once between two parallel lines like this, the angle at the bend is the sum of the angles the path makes with the two lines.




**Example 2.**
Triangle $ABC$ has $AB = AC$ and $\angle BAC = 40°$. The point $D$ lies on $AC$ with $BD = BC$. Find $\angle ABD$.

*Solution.*
Only one angle of triangle $ABD$ is known, so we start in triangle $ABC$. It is isosceles with apex angle $40°$, so its base angles are $\angle ABC = \angle ACB = \frac{1}{2}(180° - 40°) = 70°$.
Triangle $BCD$ has $BD = BC$, so its angles at $D$ and $C$ are equal. Since $D$ lies on $AC$, $\angle BCD$ is the same angle as $\angle ACB$; this links the two triangles. So $\angle BDC = \angle BCD = 70°$, and $\angle DBC = 180° - 140° = 40°$. Therefore $\angle ABD = 70° - 40° = 30°$.
Check: $\angle BDC$ is the exterior angle of triangle $ABD$ at $D$, so it should equal $40° + 30° = 70°$, and it does.

## 1.4 Polygons

**Fact 4 (polygons).**
The interior angles of an $n$-sided polygon add to $(n - 2) \times 180°$. The exterior angles of a convex polygon add to $360°$.

A polygon is *convex* if each interior angle is less than $180°$. For the first part, join one vertex to every other vertex. This cuts a convex $n$-sided polygon into $n - 2$ triangles (one for each side not touching that vertex), whose angles together make up the angles of the polygon. The *exterior angle* at a vertex is the angle between one side and the next side extended; it lies on a straight line with



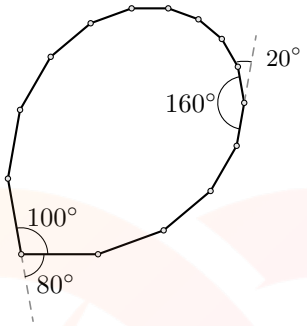
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the interior angle, so the two add to $180^\circ$. For the second part, walk once round the boundary. At each corner you turn through the exterior angle, and you finish facing the way you started, having made one full turn.

A regular polygon has all its sides equal and all its angles equal, so

$$\text{exterior angle} = \frac{360^\circ}{n}, \qquad \text{interior angle} = 180^\circ - \frac{360^\circ}{n}.$$

Know the interior angles $60^\circ, 90^\circ, 108^\circ, 120^\circ, 135^\circ, 144^\circ, 150^\circ$ for $n = 3, 4, 5, 6, 8, 10, 12$. With regular polygons, work with the exterior angle first, since it is a single division.

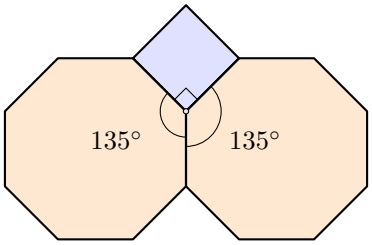



**Example 3.**
Every interior angle of a convex polygon is $160^\circ$, except for one angle, which is $100^\circ$. How many sides does the polygon have?

*Solution.*
Interior angles would give the equation $(n-1) \times 160^\circ + 100^\circ = (n-2) \times 180^\circ$, but exterior angles are quicker. The exterior angle is $20^\circ$ at each ordinary corner and $80^\circ$ at the special one. The exterior angles add to $360^\circ$, so the $20^\circ$ angles add to $280^\circ$, and there are $280 \div 20 = 14$ of them. The polygon has 15 sides.
Check: $14 \times 160^\circ + 100^\circ = 2340^\circ$, and $(15-2) \times 180^\circ = 2340^\circ$.

## 1.5 Tilings

Regular polygons fit together round a point, with no gaps and no overlaps, exactly when their angles there add to $360^\circ$. Copies of one regular polygon can do this only if its interior angle divides $360^\circ$, as for triangles ($6 \times 60^\circ$), squares ($4 \times 90^\circ$) and hexagons ($3 \times 120^\circ$). Three regular pentagons leave a gap of $36^\circ$. A regular polygon with more than six sides has interior angle between $120^\circ$ and $180^\circ$, so two copies leave a gap and three overlap. Hence triangles, squares and hexagons are the only regular polygons that tile on their own.



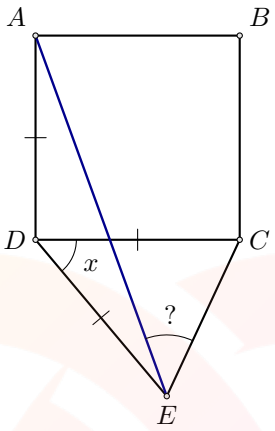



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Different regular polygons can share a point: a square and two regular octagons, as shown, since $90^\circ + 135^\circ + 135^\circ = 360^\circ$, or an equilateral triangle and two regular 12-gons, since $60^\circ + 150^\circ + 150^\circ = 360^\circ$.

### 1.6 When an angle cannot be found directly

If the information does not fix every angle in the diagram, call one unknown angle $x$, write the others in terms of $x$, and chase. Either $x$ cancels, or the question needs more information.




**Example 4.**
$ABCD$ is a square, and $E$ is a point on the other side of the line $CD$ from $A$, such that $DE = DC$ and $\angle CDE$ is acute. Find $\angle AEC$.

*Solution.*
Nothing in the question fixes $\angle CDE$, so we call it $x$ and expect it to cancel. The equal lengths are $DA = DC = DE$. Triangle $DCE$ is isosceles with apex angle $x$, so $\angle DEC = \frac{1}{2}(180^\circ - x) = 90^\circ - \frac{1}{2}x$. Triangle $DAE$ is isosceles with apex angle $\angle ADE = 90^\circ + x$, so $\angle DEA = \frac{1}{2}(90^\circ - x) = 45^\circ - \frac{1}{2}x$. From the diagram,

$$\angle AEC = \angle DEC - \angle DEA = (90^\circ - \tfrac{1}{2}x) - (45^\circ - \tfrac{1}{2}x) = 45^\circ.$$

In a multiple-choice question, choosing $x = 60^\circ$ (which makes $DCE$ equilateral) gives a quick check, but only the argument with $x$ shows that every allowed position of $E$ gives $45^\circ$.

**Common traps.**

- Trusting the picture. Lines are parallel, or sides equal, only if the question says so or you can prove it.
- Mixing up interior and exterior angles, or treating the sum $(n-2) \times 180^\circ$ as a single angle.
- Forgetting the second case: an isosceles triangle with an angle of $50^\circ$ could have it at the apex or at the base.



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### 1.7 How to spot this in a contest

1. If regular polygons appear, write down their interior angles first, and mark every side equal to the side they share.
2. Look for isosceles triangles before anything else.
3. If a path bends between parallel lines, draw an extra parallel line through the bend.
4. If an angle seems impossible to find, call one unknown angle $x$ and expect it to cancel.
5. If the question asks for the number of sides of a polygon, use exterior angles.
6. Before answering, check that the angles of each triangle you used add to $180^\circ$.

## 2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

**Problem 1** $\star$ Each interior angle of a regular polygon is $140^\circ$. How many sides does it have?

**Problem 2** $\star$ $ABCD$ is a trapezium with $AB$ parallel to $DC$. The sides $AD$ and $DC$ are equal, and $\angle ADC = 110^\circ$. Find $\angle CAB$.

**Problem 3** $\star\star$ In triangle $ABC$, the point $D$ is the midpoint of $BC$, and $DA = DB$. Explain why $\angle BAC = 90^\circ$.

**Problem 4** $\star\star$ A square, a regular pentagon and a regular $n$-sided polygon, all with the same side length, fit together exactly round a point, with no gaps and no overlaps. Find $n$.

**Problem 5** $\star\star\star$ Explain why a convex polygon can have at most three acute interior angles.

**Problem 6** $\star\star\star$ $A$, $B$ and $C$ are three consecutive vertices of a regular polygon. An equilateral triangle $ABP$ is drawn inside the polygon. Given that $\angle BPC = 40^\circ$, how many sides does the polygon have?

## 3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT JMC 2015 Question 6** $\star$
   *Find the angles inside the triangle from the angles outside it.*
2. **AMC 8 2017 Problem 6** $\star$
   *Write the three angles as multiples of one unknown.*
3. **UKMT JMC 2018 Question 3** $\star$
   *What angle does a diagonal of a square make with its sides?*
4. **UKMT Junior Kangaroo 2018 Question 7** $\star\star$
   *Two isosceles triangles: call one base angle $x$.*



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5. **AMC 8 2000 Problem 13** ⋆⋆
   *Base angles of the isosceles triangle first, then the bisector.*

6. **UKMT Junior Kangaroo 2017 Question 13** ⋆⋆
   *Co-interior angles along each of the two sides that are not parallel.*

7. **UKMT JMC 2017 Question 11** ⋆⋆
   *Add up all the angles at all seven vertices together.*

8. **UKMT JMC 2019 Question 21** ⋆⋆
   *Sides of different polygons are equal: look for an isosceles triangle.*

9. **AMC 8 2016 Problem 23** ⋆⋆⋆
   *Join E to both centres; all the radii are equal.*

10. **UKMT JMC 2015 Question 25** ⋆⋆⋆
    *Exterior angles of two triangles, then the isosceles triangle.*

11. **UKMT JMC 2018 Question 25** ⋆⋆⋆
    *Find the angle of the polygon from the angle sum of the quadrilateral.*

12. **UKMT JMC 2019 Question 24** ⋆⋆⋆
    *Look for two congruent triangles and equal exterior angles.*



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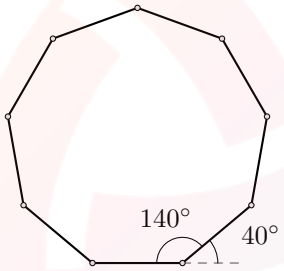
## 4 Hints

1. Find the exterior angle first.
2. Which triangle has two equal sides? After that, use the parallel lines.
3. There are two isosceles triangles. Call their base angles $x$ and $y$.
4. The three angles at the point add to $360°$. Find the third interior angle, then its exterior angle.
5. Look at the exterior angles. What can you say about the exterior angle at an acute corner?
6. Which two sides of triangle $PBC$ are equal?

## 5 Solutions

### Problem 1

*Each interior angle of a regular polygon is $140°$. How many sides does it have?*



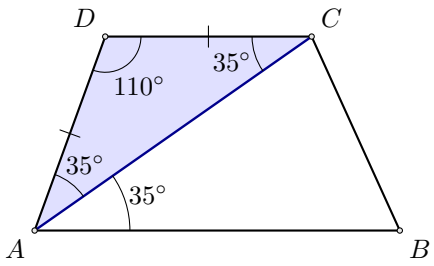

At each vertex the interior angle and the exterior angle lie on a straight line, so each exterior angle is $180° - 140° = 40°$. The exterior angles of a polygon add to $360°$, and in a regular polygon they are all equal, so there are $360 \div 40 = 9$ of them. The polygon has 9 sides.

As a check, the interior angles add to $9 \times 140° = 1260°$, and $(9 - 2) \times 180° = 1260°$.

**Answer:** 9

### Problem 2

*$ABCD$ is a trapezium with $AB$ parallel to $DC$. The sides $AD$ and $DC$ are equal, and $\angle ADC = 110°$. Find $\angle CAB$.*






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Since $DA = DC$, triangle $ADC$ is isosceles, and its base angles are $\angle DAC = \angle DCA = \frac{1}{2}(180^\circ - 110^\circ) = 35^\circ$.

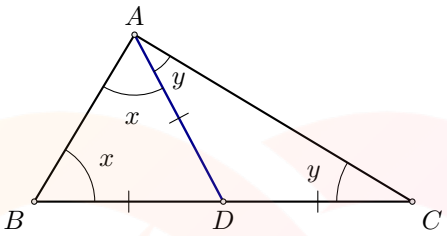
The diagonal $AC$ crosses the parallel lines $AB$ and $DC$, and $\angle CAB$ and $\angle DCA$ are alternate angles (the Z shape). Hence $\angle CAB = \angle DCA = 35^\circ$.

Note that $AC$ bisects $\angle DAB$, since both halves are $35^\circ$. The same argument shows this in every trapezium with $AD = DC$.

**Answer:** $35^\circ$

### Problem 3

*In triangle $ABC$, the point $D$ is the midpoint of $BC$, and $DA = DB$. Explain why $\angle BAC = 90^\circ$.*



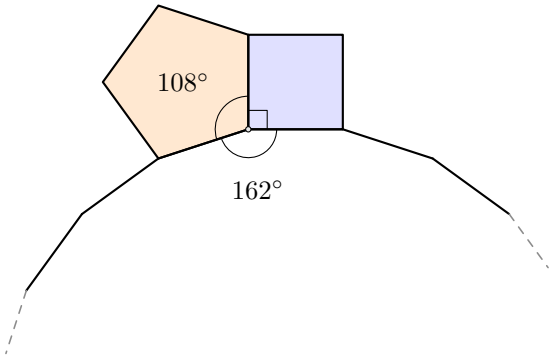

Since $D$ is the midpoint of $BC$, we have $DB = DC$, and so $DA = DB = DC$. Triangle $ABD$ is isosceles with $DA = DB$, so its base angles are equal; call them $x$, so that $\angle DBA = \angle BAD = x$. Triangle $ADC$ is isosceles with $DA = DC$; call its base angles $y$, so that $\angle DCA = \angle CAD = y$.

The angles of triangle $ABC$ are $\angle B = x$, $\angle C = y$ and $\angle A = \angle BAD + \angle CAD = x + y$. They add to $180^\circ$, so $2x + 2y = 180^\circ$, which gives $x + y = 90^\circ$. Hence $\angle BAC = x + y = 90^\circ$.

This fact is worth remembering: if the midpoint of one side of a triangle is the same distance from all three vertices, the angle opposite that side is a right angle. At Level 2 it reappears as the theorem that the angle in a semicircle is a right angle.

### Problem 4

*A square, a regular pentagon and a regular $n$-sided polygon, all with the same side length, fit together exactly round a point, with no gaps and no overlaps. Find $n$.*




The angles at the point add to $360^\circ$. The square contributes $90^\circ$. The interior angles of a pentagon add to $3 \times 180^\circ = 540^\circ$, so each angle of the regular pentagon is $540^\circ \div 5 = 108^\circ$. Hence the third polygon has interior angle $360^\circ - 90^\circ - 108^\circ = 162^\circ$.



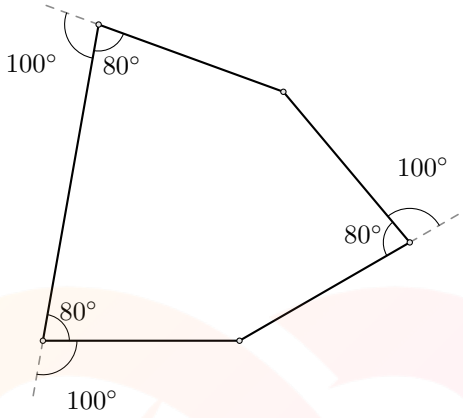
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Its exterior angle is $180° - 162° = 18°$, and the exterior angles add to $360°$, so it has $360 \div 18 = 20$ sides.

**Answer:** 20

### Problem 5

*Explain why a convex polygon can have at most three acute interior angles.*



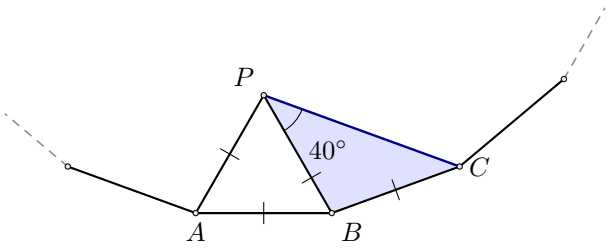

At each vertex, the interior and exterior angles add to $180°$. So if the interior angle is acute (less than $90°$), the exterior angle is more than $90°$. In a convex polygon every exterior angle is positive, and together they add to $360°$.

Suppose the polygon had four or more acute angles. The exterior angles at four of them would add to more than $4 \times 90° = 360°$, and all the other exterior angles are positive, so the total of the exterior angles would be more than $360°$. This contradicts Fact 4. Hence there are at most three acute angles.

Three is possible: an equilateral triangle has three acute angles, and so does the pentagon in the diagram, whose angles are $80°$, $150°$, $80°$, $150°$ and $80°$. (They add to $540° = 3 \times 180°$, as they should.)

### Problem 6

*$A$, $B$ and $C$ are three consecutive vertices of a regular polygon. An equilateral triangle $ABP$ is drawn inside the polygon. Given that $\angle BPC = 40°$, how many sides does the polygon have?*




The sides $BP$ and $BA$ belong to the equilateral triangle, and $BA$ and $BC$ are sides of the regular polygon, so $BP = BA = BC$. Triangle $PBC$ is therefore isosceles with $BP = BC$, and its base angles are equal: $\angle BCP = \angle BPC = 40°$. Its apex angle is $\angle PBC = 180° - 40° - 40° = 100°$.



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The interior angle of the polygon at $B$ is made up of $\angle ABP = 60^\circ$, an angle of the equilateral triangle, and $\angle PBC = 100^\circ$, so it is $160^\circ$. The exterior angle is $20^\circ$, so the polygon has $360 \div 20 = 18$ sides.

As a check, in a regular 18-gon the interior angle is $180^\circ - 20^\circ = 160^\circ$, so $\angle PBC = 160^\circ - 60^\circ = 100^\circ$, and the base angles of triangle $PBC$ are $\frac{1}{2}(180^\circ - 100^\circ) = 40^\circ$.

**Answer:** 18